

## Assignment #5(GCCF)

### SECTION A (2\*4=8)

Q1. (CO5) What is Dataflow?

Ans: Dataflow is a Google Cloud service that provides unified stream and batch data processing at scale. It enables users to create data pipelines that read from one or more sources, transform the data, and write the data to a destination. Dataflow supports both streaming and batch processing, allowing users to process fluctuating volumes of real-time data with low latency. The service is built on the open-source Apache Beam project and uses the same programming model for both batch and stream analytics.

Q2. (CO5) What is Dataprep?

Ans: Dataprep refers to **Google Cloud Dataprep by Trifacta**, a service offered on Google Cloud Platform (GCP) [cloud.google.com]. It's an intelligent data preparation tool that allows users to visually explore, clean, and transform data for analytics and machine learning tasks .

Dataprep is a low-code data preparation library in Python designed for data scientists and analysts. It is optimized for use in computational notebooks and supports both batch and stream data processing. Dataprep provides a unified interface for cleaning and standardizing data, as well as exploratory data analysis (EDA) and data visualization. The library is built on top of Pandas and Dask DataFrames and can be seamlessly integrated with other Python libraries.

Q3. (CO5) Differentiate between Cloud Tracing and Monitoring.

Ans: Cloud tracing and monitoring are both essential tools for understanding and maintaining cloud-based applications, but they serve different purposes:

#### **Cloud Monitoring**

- **Focus:** Overall health and performance of an application.
- **Data:** Tracks metrics like CPU usage, memory consumption, network traffic, response times, error rates, etc.

- **Goal:** Proactive identification of potential issues and ensuring smooth application operation.
- **Analogy:** Monitoring your car's dashboard to see if the engine is running hot, if there's enough gas, or if the tire pressure is low.
- **Tools:** Prometheus, Nagios, Datadog, etc.

### Cloud Tracing

- **Focus:** Following a specific user request or transaction as it travels through different parts of a complex application.
- **Data:** Captures detailed information about each step of a request, including timestamps, function calls, and data passed between services.
- **Goal:** Diagnose performance bottlenecks, identify errors within specific transactions, and understand how data flows through the system.
- **Analogy:** Following a package on a delivery journey, tracking each stop it makes and identifying any delays or issues.
- **Tools:** Open tracing, Zipkin, Jaeger, etc.

Here's a table summarizing the key differences:

| Feature | Cloud Monitoring                                    | Cloud Tracing                                               |
|---------|-----------------------------------------------------|-------------------------------------------------------------|
| Focus   | Application health and performance                  | Specific user request or transaction                        |
| Data    | Metrics (CPU, memory, network, etc.)                | Request timestamps, function calls, data flow               |
| Goal    | Proactive issue identification and smooth operation | Diagnose bottlenecks, identify errors, understand data flow |
| Analogy | Car dashboard                                       | Package delivery journey                                    |

### Q4. (C05) Explain Dataproc.

Ans: Dataproc in Google Cloud is a fully managed cloud service designed for running Apache Spark and Apache Hadoop clusters efficiently and cost-effectively. It supports various open-source tools and frameworks, including Apache Flink and Presto, enabling tasks like data lake modernization, ETL

processes, and secure data science at scale. Dataproc offers benefits such as serverless deployment, enterprise-grade security, and cost-effectiveness, with features like serverless deployment, advanced security configurations, and reduced total cost of ownership compared to on-premises solutions. It integrates seamlessly with other Google Cloud services, allowing data scientists and engineers to build and train models faster and access data applications easily.

## # SECTION B (6\*2=12 Mark)

Q1. (CO5) Explain the architectural design of compute and storage clouds.

**Ans:** The architecture of compute and storage clouds can be generally divided into two main parts: the **frontend** and the **backend**.

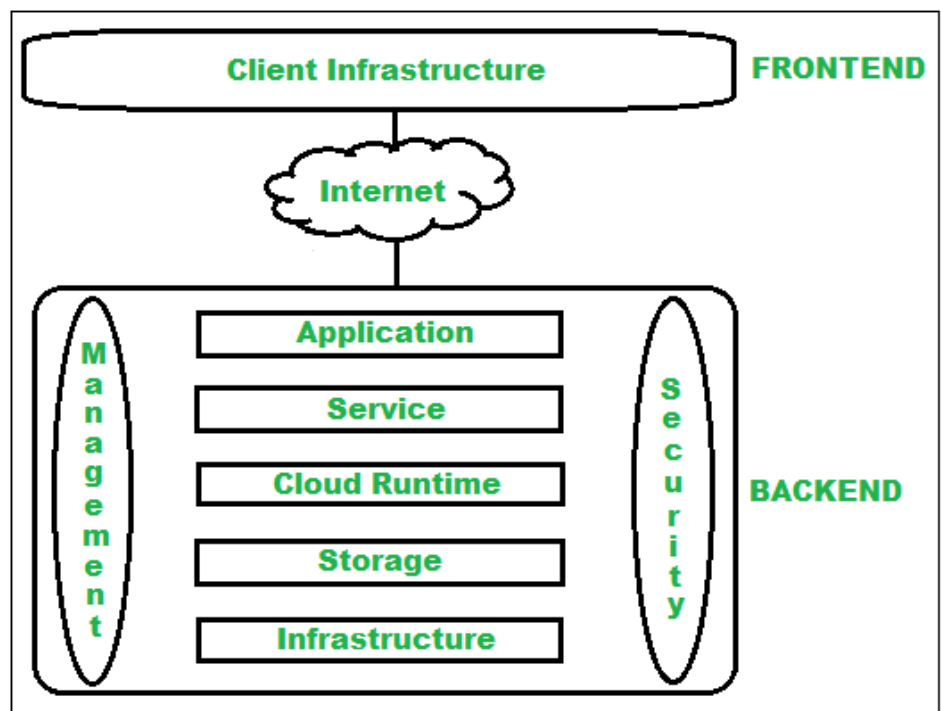
### **Frontend (Client Side):**

- This is the part that users interact with. It includes:
  - User interfaces: This could be a web browser, a mobile app, or any other interface that allows users to access cloud services.
  - Applications: These are the software programs that users run on the cloud.

### **Backend (Cloud Provider Side):**

- This is the heart of the cloud system and consists of several key components:
  - **Applications:** These are pre-configured applications offered by the cloud provider, like productivity suites or CRM software (SaaS - Software as a Service).
  - **Services:** These are building blocks for applications, offering functionalities like databases, security, or machine learning (PaaS - Platform as a Service).
  - **Runtime cloud:** This layer manages the execution of applications and services. It includes tools for provisioning resources, containerization (like Docker), and job scheduling.
  - **Storage:** This refers to the physical infrastructure that stores data, including hard disk drives, solid-state drives, and object storage systems.

- **Infrastructure:** This layer consists of the physical hardware resources like servers, network devices, and virtualization software. Virtualization allows for efficient resource utilization by creating multiple virtual machines on a single physical server.
- **Management:** This layer takes care of provisioning resources, monitoring system health, and ensuring security.
- **Security:** This layer includes firewalls, intrusion detection systems, and other tools to protect user data and applications from unauthorized access.

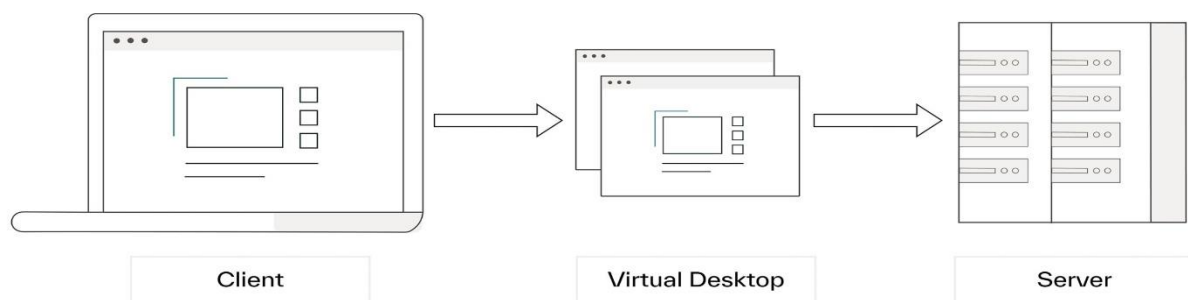


By utilizing these components and service models, cloud providers offer a scalable and elastic environment for compute and storage needs, accessible from anywhere with an internet connection.

Q2. (C05) Write short note on i) Desktop virtualization ii) Server virtualization.

Ans: (i).Desktop Virtualization:

Desktop virtualization is a technology that separates the desktop environment and associated application software from the physical client device. It allows users to access their desktop virtually from any location using different devices. The main benefits of desktop virtualization include user mobility, portability, and easy management of software installation, updates, and patches. It can be implemented locally or remotely, and various methods such as terminal server virtualization, OS streaming, virtual desktop infrastructure (VDI), and desktop as a service (DaaS) are available.



- **Benefits:**

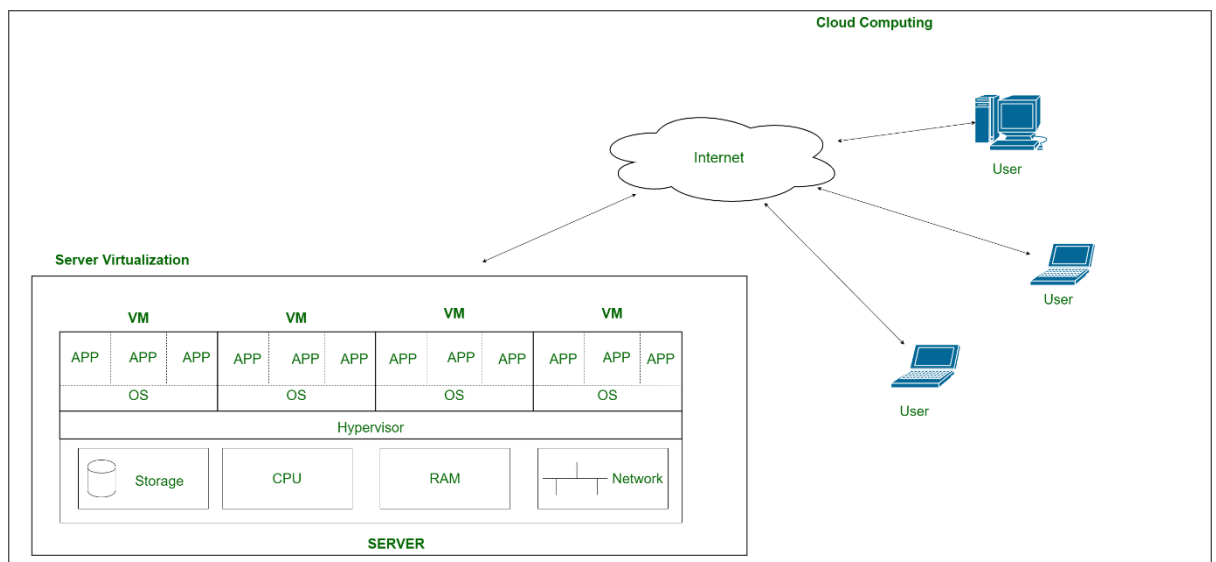
- Improved manageability: IT can centrally manage and update desktops for all users.
- Increased security: Data resides on the server, enhancing security and simplifying disaster recovery.
- Device flexibility: Users can access their desktops from any device with the appropriate software.

- **Drawbacks:**

- Reliance on network connectivity: Users require a stable internet connection for smooth operation.
- Hardware dependency: The server needs sufficient resources to support multiple virtual desktops.

## (ii). Server Virtualization:

Server virtualization is the partitioning of a physical server into multiple virtual servers. It allows multiple servers to run on a single physical machine, increasing efficiency and reducing costs. Server virtualization is commonly used for workloads such as databases, file sharing, graphics virtualization, and media delivery. It provides benefits like increased performance, reduced energy consumption, and reduced infrastructural costs. Server virtualization is typically used in environments where servers perform similar functions day-to-day, requiring specialized software and tools for management.



## • Benefits:

- Improved resource utilization: A single physical server can run multiple virtual servers, maximizing hardware usage.
- Increased agility: Virtual servers can be easily provisioned, migrated, and scaled based on demand.
- Reduced costs: Server virtualization helps consolidate hardware, leading to lower energy consumption and maintenance costs.

## • Drawbacks:

- Requires expertise: Setting up and managing virtual servers requires technical knowledge.
- Potential performance overhead: Virtualization software adds a layer of abstraction, which can impact performance in some cases.

## SECTION D (10\*2=20Mark)

### Q1. (CO5) What are the challenges in SaaS? Explain

**Ans:** The main challenges in SaaS (Software as a Service) can be categorized into the following areas:

#### ## Product Challenges

1. Providing a differentiated product or service to stand out in a crowded market.
2. Ensuring a great product experience by balancing features and simplicity.
3. Educating users on the product's capabilities and value.
4. Leveraging product analytics to drive user engagement and retention.

#### ## Customer Challenges

1. Delivering excellent customer support at scale, which can be addressed by using chatbots for common inquiries.
2. Engaging customers regularly through personalized communications and content.
3. Offering personalized experiences based on customer data and behaviour.
4. Providing value-added services like training and consulting to help customers get the most out of the product.

5. Rewarding customer loyalty with discounts, exclusive offers, and personalized rewards.

## ## Technical Challenges

1. Building a robust multi-tenancy architecture to serve multiple customers efficiently.
2. Enhancing scalability and performance to handle growing user and data volumes.
3. Ensuring data security and compliance with industry regulations.
4. Enabling seamless integration with other SaaS applications and on-premises systems.
5. Delivering regular updates to introduce new features, fix bugs, and improve security.

## ## Business Challenges

1. Hiring and managing a high-performing team in a hybrid or dispersed work environment.
2. Automating processes and streamlining operations to cut costs and improve efficiency.
3. Navigating the competitive landscape and finding the right product-market fit.
4. Securing funding and managing resources effectively to support growth.

To overcome these challenges, SaaS companies need to adopt a strategic approach focused on sustainable growth, customer satisfaction, and continuous innovation. This includes investing in product development,



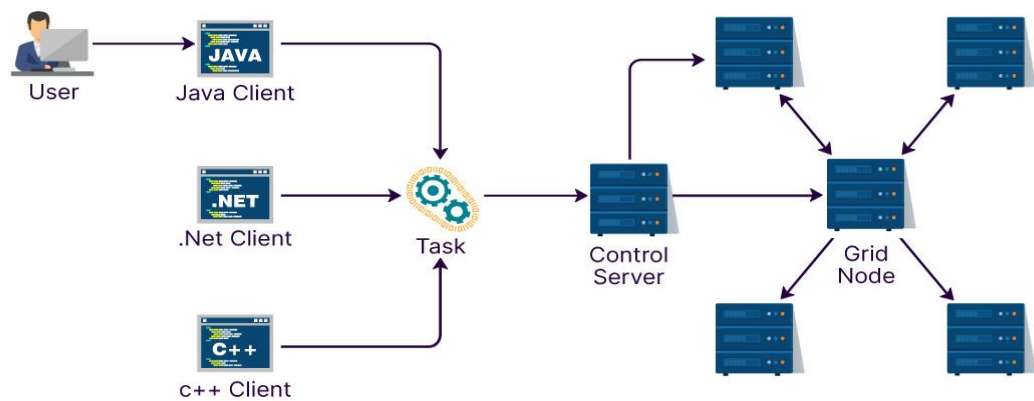
customer success, and data security, while also optimizing business operations and adapting to market changes.

**Q2. (CO5) How grid computing and utility computing are differed with each other? Explain with proper definition and major points in support of difference.**

**Ans:** Grid computing and utility computing, while both related to shared computing resources, have distinct purposes and functionalities. Here's a breakdown of their key differences:

### **Grid Computing**

- **Definition:** Grid computing leverages unused processing power from geographically dispersed computers to tackle large, complex problems that a single computer cannot handle efficiently. These computers can belong to different organizations or individuals and are connected through a network to form a "virtual supercomputer."
- **Focus:** Grid computing is primarily concerned with **sharing computing resources** to solve computationally intensive tasks in scientific research, weather forecasting, financial modeling, etc.
- **Ownership and Control:** Grids can involve resources from various administrative domains, requiring collaboration and agreements between participating entities. There might not be a central ownership or control structure.
- **Billing Model:** Grid computing often doesn't involve direct billing for resources. Participation might be based on reciprocity or collaborative research goals.
- **Accessibility:** Setting up and managing grid infrastructures can be complex, requiring technical expertise.

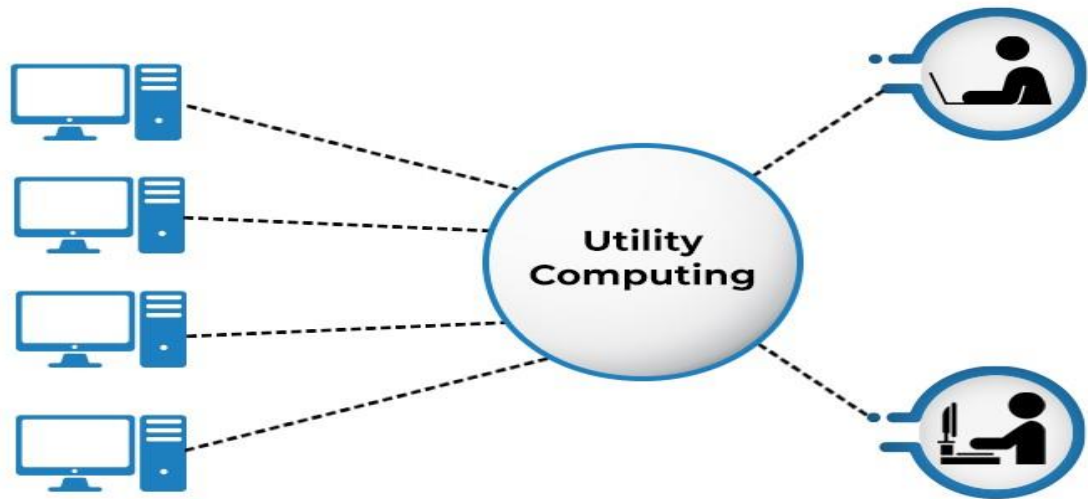


HOW GRID COMPUTING WORKS

## Utility Computing

- **Definition:** Utility computing is a service model that provides computing resources (processing power, storage, software) on-demand, similar to how utilities like electricity are delivered. Users pay only for the resources they use, similar to a pay-as-you-go pricing model.
- **Focus:** Utility computing focuses on **delivering computing resources** as a service, making them readily available and scalable based on user needs. It's often used for business applications like email, web hosting, or enterprise resource planning (ERP).
- **Ownership and Control:** A cloud service provider (CSP) typically owns and manages the infrastructure in utility computing. Users access resources through a self-service portal or API.
- **Billing Model:** Utility computing follows a **pay-per-use** billing model, where users are charged based on their resource consumption (CPU, storage, bandwidth).
- **Accessibility:** Utility computing services are designed to be user-friendly and accessible with minimal technical knowledge required.

# Utility Computing



Here's a table summarizing the key points:

| Feature           | Grid Computing                                  | Utility Computing                                   |
|-------------------|-------------------------------------------------|-----------------------------------------------------|
| Definition        | Shares unused computing power for complex tasks | Delivers computing resources on-demand              |
| Focus             | Sharing resources                               | Delivering resources as a service                   |
| Ownership/Control | Varied, often collaborative                     | Owned and managed by a cloud service provider (CSP) |
| Billing Model     | Not always direct billing                       | Pay-per-use model                                   |
| Accessibility     | Complex, requires technical expertise           | User-friendly, minimal technical knowledge needed   |

In essence, grid computing focuses on harnessing underutilized resources for specific large-scale tasks, while utility computing provides a readily available, scalable computing environment for various applications.

## Difference between grid computing and utility computing

This slide represents the difference between grid computing and utility computing based on factors such as architecture, working, tasks performed, focus, types of both, and their real-world applications.

|                                                                                     | Factors       | Grid computing                                                                                                                                                  | Utility computing                                                                                                                        |
|-------------------------------------------------------------------------------------|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
|    | Architecture  | Process architecture that unites various computer resources from many places to achieve a shared purpose                                                        | Process architecture that provides pay-per-use computing resources and infrastructure on demand                                          |
|    | Working       | Divides work across numerous systems and allow computers to contribute their particular resources to a shared objective                                         | Enables organizations to assign and divide computer resources and infrastructure to different users according to their needs             |
|    | Tasks perform | Makes greater use of current resources, responds to rapidly changing consumer demands, improves computing abilities, and offers flexibility, among other things | Lowers IT costs, makes management easier, increases flexibility, compatibility, and convenience                                          |
|    | Focus         | Primarily concerned with the sharing of computing resources                                                                                                     | Primarily concerned with obtaining computing resources                                                                                   |
|    | Types         | Computational grids, data grids, and collaborative grids                                                                                                        | Internal and external utility                                                                                                            |
|  | Applications  | ATMs, back-end infrastructures, marketing research, and more                                                                                                    | Large corporations, such as Amazon, Google, and others, use it to create their utility services for computing, storage, and applications |
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